

OpsGate

Technical Documentation

Architecture, authentication, tool reference, and security model
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OpsGate is a multi-tenant MCP (Model Context Protocol) server that governs how AI coding assistants — Claude and Replit Agents — are allowed to plan and implement changes on a real codebase. It compiles plain-language requests into self-contained, gate-checked implementation prompts, enforces deterministic scope/capability/path rules before any work happens, and carries state across Replit's disposable per-session model so a multi-phase task can resume correctly. This document is the complete technical reference: architecture, the tenant/authentication model, the full tool inventory, the governed prompt-compiler chain, and the security properties this system actually guarantees (and where it deliberately does not).

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1. System Overview

- **Claude** — turns a plain-language request into a routed, gate-checked, self-contained implementation prompt.
- **A Replit Agent** — executes that prompt inside the actual project and reports back what it did.
- **Neither agent defines or enforces the rules itself** — both check against the same OpsGate server, so the rules exist in exactly one place.

Why the split exists

- **A Replit session has no memory between prompts** — each one is a fresh session with no knowledge of any earlier one.
- **Claude tracks state across that gap** — using OpsGate's own persisted run state (Section 4), not its own conversation memory, which would not survive a new Replit session either.

What OpsGate is not

- **It does not write code.** — Every tool it exposes is a deterministic check, a text-compilation step, or a state read/write — never an implementation action.
- **It is not a general-purpose MCP framework.** — It is one purpose-built server for one governance problem: keeping AI-assisted changes to a real codebase inside agreed, auditable boundaries.
- **The two-mount split (Section 2) is a discoverability convenience, not a security boundary.** — Both mounts share identical authentication and tenant resolution; the gates inside each tool call are what actually enforce anything.

Multi-tenancy

- **OpsGate is one hosted service multiple projects (“tenants”) connect to** — not one deployment per project.
- **Each tenant has its own resolved profile** — frontend/backend write roots, extra protected paths, an optional business-context file reference, and its own credentials.
- **A tenant can never see or affect another tenant's profile, tokens, run state, or audit history.**
- **Every tenant-scoped function resolves its tenant identity purely from the caller's own authenticated token** — never from a caller-suppliable field.

2. Architecture

One Python process runs a single Starlette ASGI application with two mounted sub-applications, each its own FastMCP instance exposing only the tools that role needs.

Request flow

- **Claude and a Replit Agent each connect over HTTPS with a token** — to the OpsGate MCP server (one process, one Starlette app).
- **Claude's calls go through /mcp/claude/** — Replit's go through /mcp/replit/ — both pass through the identical `TokenAuthMiddleware`.
- **Every call resolves tenant identity against tenants/registry.json** — reads/writes run state and decisions under `runs/<tenant_id>/`, and reads knowledge resources from `content/**`.

Two mounts, one auth boundary

- **/mcp/claude/ exposes 25 tools** — 14 shared, 11 exclusive to the prompt-compiler chain and run recovery.
- **/mcp/replit/ exposes 16 tools** — the same 14 shared tools, plus 2 exclusive to instruction sync.
- **Both sit behind the identical TokenAuthMiddleware** — a valid token authenticates identically against either path.
- **The split exists purely so each caller sees a shorter, more accurate tool list for its own role** — it is explicitly documented in code as not a security boundary.

Stateless HTTP transport

- **Both FastMCP instances are constructed with `stateless_http=True`.**
- **Every incoming request gets its own fresh transport and task** — with no persistent `Mcp-Session-Id` concept at all.
- **This is a deliberate, security-motivated choice, not a default** — see Section 9 for why stateful mode was actively unsafe here.

Where state actually lives

- **Nothing needed across calls lives in server memory.**
- **Everything persists to disk under OpsGate's own repository:** — the tenant registry (`tenants/registry.json`), per-tenant run state and logs (`runs/<tenant_id>/`), and the read-only `instruction/spec` content every tenant shares (`content/**`).
- **This is what allows the stateless-HTTP transport to cost nothing functionally** — no tool in this system ever depended on in-memory session continuity to begin with.

3. Tenant & Authentication Model

Tenant registry

- **tools/opsgate_tenants.py is the single profile-resolution mechanism** — the CLI and the MCP server both resolve through it, with no per-project hardcoded profile and no external config file to walk.
- **Storage is a file-backed JSON registry** — `tenants/registry.json`, gitignored, self-healing to 0600/0700 permissions on every save, exclusively locked via a companion `.lock` file for every read-modify-write

cycle.

- **A caller with no specific tenant identity resolves to local-dev** — a built-in default with unset write roots rather than a guessed one.

Tokens

- **Generated via secrets.token_urlsafes(32)** — only the SHA-256 hash is ever stored, never the plaintext — a leaked registry file does not itself leak usable credentials.
- **Every hash comparison in the resolution and revocation paths uses hmac.compare_digest** — not a plain equality check.
- **A token carries an admin boolean and an optional free-text label** — operator-facing only, never used for authorization — minted per-consumer so revoking one consumer's access never silently revokes another's.
- **A tenant can hold multiple tokens simultaneously** — e.g. one for Claude, one for Replit, one labeled per environment — each independently revocable.

Resolution and the admin-override path

- **resolve_tenant_from_token()** returns (tenant_id, is_admin) — for a valid, non-revoked token, or (None, False) otherwise — it always fails closed, never falling back to a default tenant.
- **A separate resolve_tenant(token, override_tenant_id) function** — supports an admin token addressing a different tenant's profile, but only when the token resolves as admin and the override ID names a real tenant; a non-admin token can never address another tenant's profile this way.
- **This override path has no caller anywhere in the live MCP server today** — documented as an open, deliberately-unused capability, not a gap in what's actually reachable.

Two authentication transports, one resolution path

- **TokenAuthMiddleware accepts either the X-Opsgate-Token header (checked first)** — or an Authorization: Bearer <token> header (the transport OAuth mandates) — both resolved through the identical tenant lookup.
- **If neither resolves to a real tenant** — the request falls back to a single shared-secret comparison (OPSGATE_MCP_TOKEN), which resolves to the local-dev identity.
- **A request matching neither path is rejected with 401** — and a WWW-Authenticate header pointing at OAuth discovery metadata.

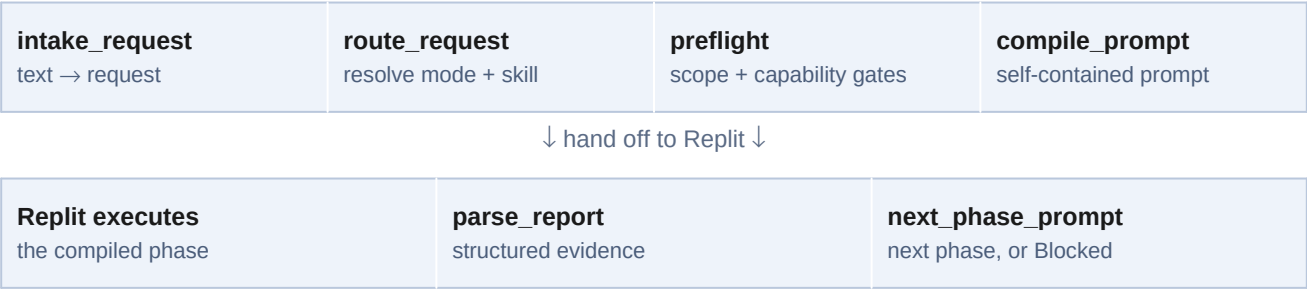
OAuth 2.1 + PKCE wrapper

- **mcp-server/opsgate_oauth.py is a minimal, purpose-built OAuth authorization server** — not a general-purpose one — added specifically because Claude's org-level Connector UI is OAuth-only with no static-header option.

- **Its /token endpoint mints nothing new** — a PKCE-verified code exchange hands back a single pre-configured backing token as the OAuth access token, so tenant resolution downstream is completely unaffected by this wrapper — it is a transport adapter, not a second identity system.
- **Covered:** — discovery metadata (/well-known/oauth-authorization-server, /well-known/oauth-protected-resource), PKCE code_challenge/verifier matching, single-use authorization codes, a pinned allowed redirect_uri, and client_id/secret verification.

4. The Governed Prompt-Compiler Chain

This is the numbered sequence Claude follows to turn a request into governed Replit work and carry the result back into the next phase. It is documented live to every connected Claude session via the opsgate://knowledge/claude-mcp-workflow resource, so the operating manual itself never drifts out of sync with the server that serves it.



next_phase_prompt's output becomes the next compile_prompt call — repeated until the run completes or a phase reports blocked.

- **1. opsgate_intake_request** — Plain-language text → a draft structured request (deliverable, outcome, module, likely authorizations), scored via a word-aware lexical matcher — not naive substring matching, so “review” inside “preview” doesn't false-match, and a genuinely ambiguous outcome is flagged rather than silently resolved to whichever deliverable happened to be checked first.
- **2. opsgate_route_request** — Resolves deliverable, internal mode, skill, required references, execution shape (bounded vs. phased), and capability — purely from the resolved routing manifest and the caller's own tenant profile, never from anything else.
- **3. opsgate_preflight** — Runs the scope, capability, and protected-path gates together, before any prompt is compiled. A missing authorization or a protected-path violation must block here — never surface later inside Replit's own session.
- **4. opsgate_compile_prompt** — Produces the literal, self-contained prompt text to hand to Replit as a new session. Every caller-supplied field is fenced or inline-sanitized before being embedded (Section 9).
- **5. opsgate_init_run** — For phased work only: persists the request, resolved route, initial gate result, and handoff state to runs/<tenant_id>/<request_id>/ — this is what survives across Replit's disposable sessions, not anything held in this conversation.
- **6. Hand off to Replit** — The compiled prompt is delivered as a new Replit session's task.

- **7. opsgate_parse_report** — Replit's final report text → structured fields: outcome, PASSED/FAILED/NOT RUN checks, files changed, HITL decisions, blockers, residual risk. A report with no recognizable structure is flagged (has_signal: false) rather than silently treated as a clean pass.
- **8. opsgate_next_phase_prompt** — Given the current run state and the parsed report, emits the next phase's prompt — or refuses outright ("Phase Blocked" / "No Next Phase") if the run's own status is blocked or the report was unparseable, regardless of what any individual phase's status looks like.

Bounded vs. phased execution

- **Bounded** — the outcome can be safely inspected, implemented, verified, and rolled back together in one batch.
- **Phased** — the request involves multiple independently reversible surfaces, schema/migration/seeding work, authentication or public-contract changes, broad refactors, or anything that cannot be verified and rolled back as one unit.
- **Only phased runs get persisted state via opsgate_init_run** — a phase prompt only ever grants authority for that one phase — prior reports are evidence, never standing authority for what comes next.

5. Capability Gates & Protected Paths

Every capability has a declared default posture and a list of preconditions it requires regardless:

Capability	Default posture	Requires (evidence needed to proceed)
ordinary_application_change	allowed_when_scoped	observable outcome, exact scope inside normal write paths
audit_or_diagnosis_without_fixes	read_only	approved inspection scope
broad_architecture_refactor	blocked	explicit broad outcome, named approved source tree, preserved contracts, phased rollback
instruction_maintenance	blocked	explicit instruction-change request, exact instruction paths
schema_migration_backfill	blocked	explicit request, approved target mapping, named paths, safe non-prod environment, rollback plan
data_seeding	blocked	explicit request, named seed paths, safe non-prod environment, approved profiles/scale, idempotency
contract_change_or_destructive_cleanup	blocked	explicit changed boundary, consumer evidence, compatibility plan, recovery plan
package_config_environment_deployment	blocked	exact explicit authorization, task-specific safety prerequisites

- **A capability whose default posture is not blocked is authorized by default** — an explicit `authorized: false` on one of those must not block it either.
- **Only a blocked-default capability requires an explicit `authorized: true` to proceed.**
- **This single rule — `capability_authorized()` in `tools/opsgate_routing.py` — is shared by every tool that makes a capability decision** — specifically so they can never disagree about the same request.

Protected paths

- **Every tenant gets a universal baseline regardless of its own configuration** — version control internals, secrets files, dependency trees, CI configuration, agent memory — merged with that tenant's own `extra_never_access` list and its resolved frontend/backend write roots.
- **A path that may resolve to protected content is treated as protected outright** — there is no partial-match exception.

6. The Human-in-the-Loop Protocol

OpsGate deliberately separates two kinds of gate failure, because only one of them is actually a decision for a human to make:

Kind	Meaning	Response
Deterministic	<code>scope_gate</code> , <code>capability_gate</code> , <code>protected_path_gate</code> — one correct answer, no judgment involved	Name the exact failed gate and what's missing. Stop. Never phrased as a decision request.
Judgment (HITL)	One of exactly three cases below — genuine ambiguity only discoverable during real work	Pause the entire task; return a structured decision request; wait for a human reply.

The three HITL cases — and only these three

- **Unknown next step** — even after bounded, approved inspection, the next step genuinely cannot be determined.
- **Tied valid options** — two materially correct answers exist and nothing in evidence, convention, or existing rules favors one.
- **Self-made scope expansion** — proceeding would require the agent to decide, on its own, to expand the approved scope.

Complexity, risk, or being security-related is explicitly not a HITL trigger on its own — if a deterministic rule already permits, requires, or forbids the action, that rule is followed directly, with no question asked.

The decision-request format

HITL decision required

ID: HITL-task-Pphase-Qnumber Blocked check: Question:

Evidence checked: Options: A. – B. –

Exact resume point: Required reply: DECIDE <A|B>

- **A human's reply is persisted outside the conversation via opsgate_record_decision** — append-only, tenant-scoped, hitl_id shape-validated against the same pattern a real HITL object requires — necessary because neither this conversation nor whichever Replit session eventually resumes the work will otherwise have any record of it.

Per-Action Gate vs. Mandatory HITL Gate

- **The full Mandatory HITL Gate evidence table runs once before editing, once before each phase, and once before the final report.**
- **A lighter Per-Action Gate** — the same three Known/Single/Bounded checks, at finer grain — runs before every individual state-changing action in between, so an agent can't drift out of bounds between checkpoints without it being caught immediately.

7. Tool Inventory

16 tools on /mcp/replit/, 25 on /mcp/claude/, 14 shared between both. Every tool resolves its tenant identity purely from the authenticated caller — never from a request field — and a deterministic gate tool returning can_proceed: false is a normal, valid result, not a protocol-level error.

Shared gate, profile, and decision tools (both mounts)

Tool	Purpose	Notes
opsgate_show_profile	Resolved tenant profile, write roots, and protected paths.	No request required
opsgate_check_capability	Standalone deterministic capability_gate check.	Shares capability_authorized()
opsgate_check_paths	Standalone deterministic protected_path_gate check.	Same matcher as preflight
opsgate_preflight	Scope + capability + protected-path gates, together.	Run before every phase
opsgate_record_decision	Appends one HITL decision to the tenant's decisions.pylog.	hitl_id shape-validated

Shared self-service and observability tools (both mounts)

Tool	Purpose	Notes
opsgate_list_own_tokens	Label/admin metadata for the caller's own tenant's tokens.	Never returns token values
opsgate_issue_own_token	Mints a new token for the caller's own tenant.	Always non-admin; shown once
opsgate_revoke_own_token	Revokes a token, only if it belongs to the caller's own tenant.	Ownership-checked
opsgate_list_audit_log	The caller's own tenant's recent tool-call history.	Filtered from a shared log file
opsgate_quota_usage	Call volume over 1h/24h/7d windows, per tool.	Visibility only, no enforcement

Shared admin-gated tools (both mounts, require an admin token)

Tool	Purpose	Notes
opsgate_admin_create_tenant	Registers a brand-new tenant.	Fails if the ID already exists
opsgate_admin_list_tenants	Every tenant's public profile, registry-wide.	Never exposes token hashes
opsgate_admin_issue_token	Mints a token for any tenant, optionally admin.	Can mint further admin tokens
opsgate_admin_revoke_token	Revokes any token, regardless of owning tenant.	No ownership check needed

Replit-only tools

Tool	Purpose	Notes
opsgate_sync_instructions	Manifest ({path, size}) of every instruction/skill file.	No content, size-capped
opsgate_sync_file	Full content for one path from that manifest.	Fetches one at a time

Claude-only tools

Tool	Purpose	Notes
opsgate_intake_request	Plain text → draft structured request.	Word-aware lexical scoring
opsgate_route_request	Resolves mode, skill, references, capability.	Pure function of request + tenant
opsgate_init_run	Persists request/route/gate/handoff for a phased run.	Writes to disk; size-capped
opsgate_list_runs	The caller's own tenant's tracked runs.	Most recent first, capped
opsgate_get_run	Full persisted state for one of the caller's own runs.	Raises on a corrupted file
opsgate_compile_prompt	The compiled prompt for a request/route pair.	Prose text, not JSON
opsgate_next_phase_prompt	Next phase's prompt, or a refusal.	Refuses on any blocked state
opsgate_parse_report	Replit's report text → structured fields.	Flags unparseable input
opsgate_lint_report	Confirms a report states every required section.	Pre-check before parsing
opsgate_lint_prompt	Confirms a compiled prompt states every required concept.	Accepts multiple phrasings
opsgate_export_ruleset	One snapshot of every knowledge resource below.	For offline/CI use

Deliberately not exposed as tools: validate-json, validate-engine, test-all — self-test commands run by hand inside this repository, not gate/routing calls a live task needs. Tenant deletion and real rate-limit enforcement are likewise deliberately unbuilt as of this writing (Section 9).

8. Knowledge Resources

Read-only governance content, backed by canonical source files read live at call time — never a copied string, so a resource can never drift out of sync with its own source.

Resource	Always-on?	Backed by
opsgate://knowledge/hitl-protocol	Yes, both mounts	The full HITL specification, unabridged
opsgate://knowledge/security-rules	Yes, both mounts	Durable security rules, scaffolding stripped

Resource	Always-on?	Backed by
opsgate://knowledge/claude-mcp-workflow	Yes, Claude only	The numbered chain in Section 4, as served live
opsgate://knowledge/skill-workflow/{skill}	Route-conditional	One skill's own numbered workflow
opsgate://knowledge/instruction-object/{name}	Route-conditional	One domain's durable rules (backend, frontend, etc.)

9. Security Model

Critical fix: stateless HTTP transport — session identity could not be trusted in stateful mode

In the MCP SDK's default stateful mode, a session's tool calls execute inside a task spawned once at session-creation time. Python's contextvars do not propagate across tasks, so a later request's own authenticated identity never reached that task — and the SDK's built-in session-owner guard never activated here, since it requires an auth provider this server does not use.

Net effect: any valid token could resume any other tenant's — or an admin's — existing session by presenting its session ID.

Verified exploitable against a real running instance, then closed by switching both FastMCP instances to `stateless_http=True`, which removes the persistent-session mechanism entirely.

A permanent regression test sends a request with a fabricated session ID and asserts it resolves strictly to the requesting token's own tenant.

Caller-supplied text is never trusted as instruction

- **Free-text fields (outcome, must_not_change, acceptance) can originate from a non-technical user's plain language via `opsgate_intake_request`** — with no review before reaching a compiled prompt.
- **Each is wrapped in a fixed delimiter pair** — `<<<CALLER_SUPPLIED_DATA>>> / <<<END_CALLER_SUPPLIED_DATA>>>`, with the delimiter itself neutralized if it appears literally inside the caller's own text, so a field cannot forge its own closing marker.
- **Inline fields embedded outside a fenced block (a title, a table cell)** — are flattened to one line and stripped of the same delimiters via `sanitize_inline_text()`, closing an injection path a prior audit found: a route with no configured capability could fall back to a raw, caller-supplied authorizations dict key, embedded unsanitized into the compiled prompt's routing table.
- **This is a real, structural mitigation, not a complete one** — no prompt-injection defense is airtight — but every hard boundary (protected paths, capability gates) is enforced by deterministic tool calls regardless of what any compiled prompt says.

Tenant isolation

- **Every tenant-scoped function resolves its tenant purely from the caller's own authenticated token** — never from a request field.
- **Independently re-verified function by function, not just asserted** — run storage, audit-log filtering, token operations, and the admin-gated tools all scope correctly, confirmed by a dedicated cross-tenant adversarial audit and by two real tenants exercised against the live running server in `tests/test_opsgate_mcp_integration.py`.

Fail-closed by construction

- **An unknown, malformed, or revoked token is rejected immediately** — never falls back to a default identity.
- **A malformed (non-object) authorization entry is treated as not-authorized** — rather than crashing or silently passing.
- **A corrupted run-state file is surfaced as a named, explicit error** — from the single-run recovery path, rather than either crashing unhelpfully or returning data indistinguishable from “never written.”
- **Run-state files are read via `ast.literal_eval`, not `exec()/import`** — a reader exposed to an MCP caller does not execute file content as code, even though the writer only ever produces safe literals today.

Bounded input, bounded output

- **Request size is capped** — 50,000 characters for `opsgate_init_run`; decision fields are capped at 5,000 characters.
- **List/log-reading tools cap the number of items a single call can return (200)** — `opsgate_list_runs`, `opsgate_list_audit_log`, regardless of what a caller asks for — including correctly treating a requested limit of exactly zero as zero results, not the default page size, and rejecting a negative limit outright rather than silently mis-slicing.

Structured, tenant-attributed audit logging

- **Every tool call — successful or not — is wrapped once at registration time** — and appended to `runs/audit.jsonl` with tenant, tool name, success, duration, and (on failure) an error message.
- **This is what actually distinguishes “which tool did this tenant call”** — from the web server's own access log, which only ever shows a generic `POST /mcp/claude/ 200 OK` regardless of which tool ran inside that request.

Deliberately not built

- **Tenant deletion over MCP was left out on purpose** — it is the single most destructive, least reversible admin operation available, and exposing it needed a deliberate decision rather than a default inclusion alongside the other admin tools.
- **Real rate-limit enforcement was also left out** — there is no existing usage-tracking baseline to set a real numeric threshold against yet, so `opsgate_quota_usage` ships as visibility only, intended to inform a

real limit once real usage data exists rather than enforcing an arbitrary one now.

10. Deployment & Operations

As of this writing, OpsGate runs as a single supervised process (not yet on dedicated cloud infrastructure) — a deliberate, acknowledged interim state, not a target architecture. Section 12 documents the recommended approach for moving off it.

Aspect	Current state
Process supervision	launchd (com.opsgate.mcpserver) — auto-restarts on crash, replacing manual process management
Public exposure	Tailscale Funnel, terminating TLS in front of the local process
Health check	Unauthenticated GET /health — confirms the tenant registry itself parses, not just that the process answers
Audit log	runs/audit.jsonl, append-only, structured per-tool-call entries
Logs	~/Library/Logs/opsgate-mcp-server.log (stdout/stderr under launchd)
Bind host	127.0.0.1 only — reachable externally only through the Funnel

- **A code change requires restarting the supervised process.**
- **A content change under content/** is picked up on the next call with no restart** — since those files are read from disk live rather than cached.
- **Every restart severs any in-flight MCP session, and a connected client must reconnect** — though with stateless HTTP now in place, there is no session to sever in the first place going forward.

11. Setup Process: Connecting a New Tenant

This section walks through standing up OpsGate for a new project end to end: provisioning the tenant, minting tokens, and connecting both the Claude and Replit sides. It draws entirely on the tenant/authentication model (Section 3), the tool inventory (Section 7), and the deployment model (Section 10) above — nothing here introduces a new mechanism.

1. Prerequisites

- **The OpsGate server is running and healthy.** — GET /health returns healthy, confirming the process is up and the tenant registry parses.
- **An admin token exists.** — Admin tokens are the only credential that can create a new tenant or mint a token on another tenant's behalf (opsgate_admin_create_tenant, opsgate_admin_issue_token).

- **The project's write boundaries are decided up front.** — Which paths are the normal working area (frontend/backend write roots), and which paths — beyond the universal protected baseline — must never be touched (`extra_never_access`).

2. Register the tenant

- **Call `opsgate_admin_create_tenant` with an admin token.** — Supply the tenant ID, the resolved write roots, any extra protected paths, and an optional business-context file reference. The call fails if the ID already exists, so it is safe to retry against a checked ID.
- **The tenant is written to `tenants/registry.json`.** — Self-healing to 0600/0700 permissions on save, exclusively locked for the write — no manual file editing is required or supported.

3. Issue tokens for each consumer

- **Mint one token per consumer, not one shared token.** — A typical tenant holds at least two: one for its Claude connection and one for its Replit connection — each independently revocable without affecting the other.
- **Use `opsgate_admin_issue_token` (admin-gated) for the initial tokens,** — or `opsgate_issue_own_token` once the tenant already holds one token and wants to self-service the rest.
- **Only the SHA-256 hash is ever stored.** — The plaintext token is shown exactly once, at mint time — record it immediately; it cannot be retrieved again, only revoked and replaced.

4. Connect the Replit side

- **Point Replit's MCP connection at the `/mcp/replit/` mount** — on the OpsGate host.
- **Authenticate with the Replit-side token** — via the `X-Opsgate-Token` header, or as an Authorization: Bearer token if the client only supports the OAuth-style transport.
- **Confirm the connection with `opsgate_show_profile`.** — A successful call confirming the expected write roots and protected paths is the signal the tenant resolved correctly — not just that the request returned 200.

5. Connect the Claude side

- **For a direct token-header client, connect at `/mcp/claude/`** — the same way as Replit, with the Claude-side token.
- **For Claude's org-level Connector UI (OAuth-only)** — use the OAuth 2.1 + PKCE wrapper (Section 3): configure the connector against the discovery metadata at `/.well-known/oauth-authorization-server`, with the pinned `redirect_uri` and client credentials for this deployment. The wrapper exchanges the OAuth flow for the same pre-configured backing token — tenant resolution is unaffected by which transport was used to present it.
- **Confirm with `opsgate_show_profile` from the Claude side as well.** — Both mounts should resolve to the same tenant profile.

6. Verify end to end

- **Run a single opsgate_preflight call for a small, ordinary_application_change request.** — A `can_proceed: true` result confirms the scope, capability, and protected-path gates are all resolving against the new tenant's real profile, not a default.
- **Check opsgate_list_audit_log** — to confirm the calls made during setup are attributed to the new tenant, not local-dev.

Ongoing maintenance

- **Rotating or revoking a token is self-service** — `opsgate_issue_own_token / opsgate_revoke_own_token` — no admin call needed once the tenant is provisioned.
- **Changing write roots or the protected-path list** — is a registry update, not a re-provisioning — it does not require new tokens.
- **A code change to the server requires a process restart; a content/** change (templates, instructions, skills) does not.** — See Section 10.

12. Recommended Approach: Migrating Off the Current Hosting

This section documents the recommended approach for moving OpsGate off a personal machine and a tunneling service onto real, durable server infrastructure. It is a plan, not a change that has been made — nothing described here has been provisioned or migrated as of this writing.

What has to be true regardless of where this runs

- **A stable public HTTPS endpoint on a real domain** — not one dependent on a specific machine being powered on and connected to the internet.
- **Persistent, non-ephemeral disk storage for tenants/registry.json and runs/**.** — This is the single most important requirement: both are plain files, not a managed database, so any hosting target whose local filesystem resets on every restart or redeploy would silently destroy the tenant registry and the entire audit/run history. Whatever target is chosen must provide a real, durable volume for this data, not container-ephemeral storage.
- **Secrets delivered as environment variables, never committed to the repository** — the shared-secret token, the OAuth client ID/secret/backing token, the allowed-hosts list, and the pinned OAuth redirect URI.
- **A process supervisor that restarts the service on crash** — the direct equivalent of the launchd configuration already in place, just running under whatever init system the target platform uses (systemd on most Linux hosts, or a platform's own process manager).
- **A loopback-only bind host behind a reverse proxy or load balancer that terminates TLS** — preserving the same shape the current 127.0.0.1-plus-Tailscale-Funnel setup already has, just with a real reverse proxy and certificate in place of the Funnel.

Three realistic hosting shapes

Option	What it looks like	Tradeoff
A. Single small VM	A cloud VM (e.g. a low-cost droplet/instance) with a persistent disk, systemd supervising the process, and a reverse proxy (e.g. Caddy, for automatic TLS) in front of it on a real domain.	Closest conceptually to the current setup — swap the Mac for a VM, launchd for systemd, the Funnel for a real proxy. Most control, simplest mental model, cheapest to reason about for a single-process, low-traffic service like this.
B. PaaS with a persistent volume	A platform that manages TLS, the domain, and deploys, but still offers an attachable persistent disk (several mainstream platforms do).	Less operational overhead than a bare VM — no systemd/proxy/certificate configuration to maintain by hand — while still satisfying the durable-storage requirement above. A reasonable middle ground.
C. Stateless/serverless platforms	Container platforms or serverless functions with only ephemeral local storage.	Not recommended without further work first. This would silently lose the tenant registry and audit/run history on every redeploy or scale event unless the storage layer is re-architected onto a real database or object store beforehand — a materially larger change than a hosting move, called out here so it is not picked by default without understanding that cost.

What changes for already-connected clients

- **Every tenant currently connected points at the current Tailscale Funnel hostname** — Replit's "Connect via MCP" URL, and any Claude org-level Connector configuration.
- **Moving to new infrastructure means updating those URLs and the OAuth connector configuration for every connected consumer** — not just standing up a new backend — this is a coordinated cutover step, not a transparent one.
- **The OAuth wrapper's pinned redirect URI and allowed-hosts configuration also need updating** — to the new domain as part of the same cutover.

Migration checklist

- Choose a hosting target from the options above and provision it with a real, persistent volume.
- Point a real domain at it and obtain TLS — automatic via a reverse proxy, or via the platform's own certificate management.
- Copy tenants/registry.json and runs/** onto the new host's persistent volume — this is the one true state migration; everything else is a fresh code deploy.
- Set the same secrets as environment variables on the new host.

- Deploy the code, start the process under the new supervisor, and verify GET /health reports healthy.
- Update every connected tenant's MCP URL and OAuth connector configuration to the new domain.
- Decommission the current Mac-hosted process only after every consumer is confirmed working against the new host — not before.

13. Content Governance (Templates)

- **Two full authoring templates define what a spec or business file must contain** — content/templates/SPEC_FILE_PROMPT_TEMPLATE.md (a 16-section implementation-spec structure) and BUSINESS_FILE_PROMPT_TEMPLATE.md (a 14-section, implementation-neutral business-file structure): document control, traceability IDs (REQ-*, NFR-*, BUS-CAP-*, BUS-REQ-*, BUS-RULE-*, DEC-*, OQ-*, REC-*), acceptance criteria, and a verification matrix.
- **Verified directly against 29 real specification files and 21 real business files from a production tenant** — both templates matched the current convention essentially verbatim, including the ID scheme.
- **The templates themselves are authoring references, not something code reads verbatim.** — compile_artifact_prompt() in tools/opsgate_prompts.py compiles a condensed equivalent for the actual prompt handed to an implementing agent.
- **A delta specification — one documenting only what changed against a prior spec — gets a structurally distinct compiled body** — a Summary-of-Changes table, per-item corrected/new/superseded sections, an Acceptance Criteria Addendum — rather than a shortened version of the full-spec body.
- **Detected from the request's own text via the phrase “delta spec”/“delta specification”** — the same phrase this system's own routing signals already use, so detection and routing agree by construction.